Kenneth Lance Howard

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Education

North Carolina A & T State University, B.S. in Computer Science- **Magna Cum Laude** **Greensboro, NC | May 2025**
North Carolina A & T State University, M.S. in Computer Science-**AI and DS focus GPA:3.91 Greensboro, NC|Exp. Dec 2026**

Technical Skills & Software Proficiencies

Proficient: Python, Java, JavaScript, React, C++, SQL, Linux, Node.js, APIs/microservices, PyTorch, Kubernetes
Skills & Interests: Full-stack Development, Mobile & Web Applications, Distributed Systems, Data Structures & Algorithms, Machine Learning, AI/ML Research, RAG, Cloud Services, Deep Learning. C# , Unity, CUDA / GPU Programming, Docker, WebSockets / real-time systems

Work Experience

Sandia National Laboratories **May 2024 – Jan 2026**
Computer Science R&D Intern – Nuclear Deterrence **Albuquerque, NM**

- Engineered a **JSON metadata pattern-matching tool** that identified anomalies in legacy national-security files, improving engineer efficiency by nearly **50%.**
- Built web-scraping and data-cleaning pipelines for the **ATLAS RAG platform**, achieving approximately **80%+ agreement** with expert-verified answers.

Graduate Research in AI/ML **Aug 2024 – Present**
AI Full-Stack Developer – EduAI **Greensboro, NC**

- Migrated a **Streamlit prototype to a React SaaS platform** with RAG pipelines and persistent user accounts.
- Built document-ingestion and vector-search pipelines; deployed EduAI to **50+ student testers**.

Boeing Research Collaboration **Aug 2025 – May 2026**
Graduate Researcher – Predictive AI Systems **Greensboro, NC**

- Generated ATC datasets and **Python pipelines** from aircraft telemetry and Unity simulations for VLM evaluation.
- Built Unity flight simulations processing **1,000+ telemetry points per run**, improving trajectory prediction by **30%**.

Projects

- **ArchAI – Master's Thesis: Agentic AI System Design Platform** (**React**, **TypeScript**, **Node.js**, **LLMs**, **MCP**, **Agents**):
  Built a full-stack SaaS platform integrating MCP servers and LLM-driven workflows to automate architecture planning, generating system blueprints, UML diagrams, and stack recommendations.
  - *Benchmarked **3 orchestration approaches** across quality, cost, latency, and consistency; conducted a blinded survey assessing UML quality across **5 criteria**.*
- **AI Data Visualization Platform – J.P. Morgan Data for Good** (**React**, **Tailwind**, **OpenAI API**, **Python**):
  Built a full-stack application that parsed and visualized NBA player statistics and mental-health trends from CSV datasets using LLM-powered analysis.
  - *Reduced analysis time by at least **50%** through automated chart generation and sentiment summaries.*
- **Capital One Custom AI Assistant – 1st Place** (**Python**, **Streamlit**, **Docker**, **LLMs**):
  Built a containerized AI assistant with session-based context retention that delivered structured and context-aware responses.
  - *Outperformed preset chatbot solutions by approximately **30%** in contextual accuracy.*
- **Deutsche Bank Hackathon – Financial Sentiment Analysis** (**Python**, **NLP**, **Real-Time Data Processing**):
  Developed a real-time pipeline that classified financial news and social-media content using NLP models integrated into a live dashboard.
  - *Achieved **90% model accuracy** (F1 = 0.91) with less than one-second latency.*

Leadership & Involvement

- **Python Bootcamp Leader, NC A&T:** Led peer-learning sessions focused on strengthening Python programming and coding fundamentals. **Spring 2024**
- **Woodland Ellroy Hall Fellowship Recipient – 1 of 15 Selected:** Recognized for academic excellence, graduate research potential, and leadership. **2025**
- **Microsoft Mentorship Program – 1 of 10 Selected:** Selected for mentorship focused on Microsoft Copilot, computer science, and professional development. **Fall 2024**
- **Dell Generative AI Certification & Award:** Earned an industry-recognized credential in generative AI and applied AI technologies. **Fall 2024**
- **Hackathon Leadership:** Earned multiple top finishes, including first-place awards at the Capital One and Chevron hackathons. **2023 – 2025**
- **Professional Organizations:** Member of ACM, NSBE, and SHPE at North Carolina A&T State University. **Fall 2023 – Present**
- **CodeHouse Frontend Engineering Bootcamp:** Selected for and completed frontend engineering training and certification. **Spring 2026**